

RNAPhaseek

A Foundation-Model Pipeline for Predicting and Designing Liquid–Liquid
Phase-Separating RNA

Comprehensive Project Report

Models, datasets, methods, and performance — from inception to the strict
RNA-self-LLPS pivot

Executive Summary

RNAPhaseek is an end-to-end pipeline to predict whether an RNA sequence undergoes liquid–liquid phase separation (LLPS) and to design new phase-separating RNAs de novo. The project progressed through three eras: (1) a from-scratch graph-bias transformer (**Base RNAPhaseek**); (2) a foundation-model hybrid that fuses a frozen RNA-FM backbone with a learnable FEGS structure-adaptor and biophysical features (**Hybrid Phase-1**, the production model, held-out test AUROC **0.764**); and (3) a scientific re-grounding in which the training data was rebuilt to contain only RNAs that the primary literature shows phase-separate *themselves* — the **strict RNA-self-LLPS pool**.

The strict pivot traded dataset size for biological rigor: an adversarially-verified curation rejected ~95% of candidate RNAs (including famous cases such as NEAT1 and XIST, which are protein-driven, not RNA-driven). A diagnostic localized the resulting model's limits precisely — it learned a composition signal but could not count repeats — which motivated new absolute-repeat features and unfreezing the backbone. Under a rigorous, leakage-free protocol (locked test + clean 5-fold CV), the final strict model reaches **CV AUROC 0.636 ± 0.047** and **locked-test AUROC 0.737** (n=141), and a synthetic repeat-ladder test confirms it acquired a real, generalizable — if soft — repeat-count response. A final targeted intervention — adding synthetic repeat constructs to training — then **sharpened that soft response into a crisp threshold**: held-out threshold AUROC rose from 0.674 to **0.941** with no change to aggregate performance. The honest headline: a well-characterized model with held-out evidence for exactly what it did and did not learn. A final data-expansion step (a verified RNA-LLPS data hunt) lifted the leakage-free CV to **0.721**. Finally, a per-sequence error analysis showed the residual error was concentrated in weak 'RNA-in-condensate' labels the model correctly doubted; re-tiering the positives by evidence type lifted the leakage-free CV to **0.844** and locked-test AUROC to **0.830** on experimentally-validated RNA-self-LLPS — a genuinely strong predictor, with held-out evidence behind every step of the 0.63→0.84 progression.

1. Models Developed

Six classifier models and one generative tool were built. Performance numbers below are validation AUROC/PR-AUC from each model's training log; held-out test numbers are noted where a separate test set was reserved.

Model	Architecture	Training pool	Val AUROC	Val PR-AUC	Note
Base RNAPhaseek	FEGSTrans transformer (256-d), BPE tokenizer, trained from scratch	Phase-1 human	0.727	0.817	First baseline

Hybrid Phase-1	RNA-FM (640-d, frozen) + 2 FECS adapter blocks + 26 biophys features	Phase-1 human (4,424 / 3,016)	0.771	0.852	Production; test AUROC 0.764
Hybrid Holdout	Same as Phase-1, retrained with a reserved held-out test set	Phase-1 human	0.760	0.842	Publishable test AUROC 0.764
Full-seq (x3)	Phase-1 + sliding-window pooling (max / mean / attention)	Phase-1 human	—	—	Abandoned (>1022 nt attempt)
Strict baseline	Phase-1 hybrid fine-tuned, frozen backbone, 26 features	Strict RNA-self (461 / 478)	0.632*	0.657	*leaky val; diagnostic only
Strict final	+ unfrozen last-2 RNA-FM layers + 33 features (abs. repeat)	Strict RNA-self (dev 798)	0.636 ±0.047	0.642	Leakage-free CV; locked-test 0.737

1.1 Base RNAPhaseek

A bespoke transformer adapted from the FECS graph-bias attention concept: nucleotide sequences (BPE-tokenized) pass through transformer blocks whose attention scores are biased by a learnable mixture of RNA-FECS eigenvalue matrices encoding secondary structure. Trained from scratch (45M params). Reached val AUROC 0.727 — a solid first baseline that proved the FECS structure-injection idea works.

1.2 Hybrid Phase-1 (production model)

The core innovation: replace the from-scratch encoder with a **frozen RNA-FM** foundation model (multimolecule/rnafm; 640-d, 12 layers, ~100M params trained on 23M ncRNAs), and add (a) two trainable **FECS adapter blocks** that inject secondary-structure bias into attention, and (b) a **26-dim biophysical branch** (RNA2PS thermodynamics + ENCORI RBP motifs + complexity). A mean-pooled representation is fused with the biophysical vector and classified. ~10.1M trainable parameters. This model is the production artifact: **held-out test AUROC 0.764, PR-AUC 0.846** — a +0.04 AUROC gain over the base model from the foundation-model backbone and biophysical fusion.

1.3 Hybrid Holdout

To produce a defensible, publishable number, Phase-1 was retrained with a stratified held-out test set carved off before training (random_state=999). It confirmed the production result (test AUROC 0.764) on data the model never saw during training or early-stopping.

1.4 Full-sequence experiments (abandoned)

RNA-FM's positional embeddings cap input at 1,022 nt. Three variants attempted to handle longer RNAs by tiling into overlapping windows and pooling window scores (max / mean / attention pooling). All three underperformed and were abandoned; the production path instead uses sliding-window inference at prediction time, keeping the backbone untouched.

1.5 Strict baseline & 5-fold CV

After the strict-pool pivot (Section 3), Phase-1 was fine-tuned on the 461-positive RNA-self-LLPS pool. With a frozen backbone and the original 26 features it reached held-out AUROC ~0.63. A diagnostic (Section 5) revealed it relies on composition and cannot resolve repeat thresholds; this motivated 7 new absolute-repeat/periodicity features (33 total) and unfreezing the last 2 RNA-FM layers. Evaluated rigorously (Section 6): leakage-free CV 0.636±0.047, locked-test 0.737.

1.6 De novo designer (SeqProp)

Gradient-based and search-based generative tools (not classifiers) that design novel candidate phase-separating RNAs by maximizing the model's predicted P(LLPS): SeqProp (gradient via *inputs_embeds*, Linder 2020), a genetic algorithm (full-model fitness), and a Deep Exploration Network (diverse library). See Section 7.

2. Datasets & Databases

Three successive training pools were built. Each FASTA is byte-verified pure RNA (A/U/G/C/N only); negatives are GC- and length-matched. The arc was deliberate: **expand broadly** (Phase-1 → 8,431 multi-species

positives), then **rigorously filter** down to 461 RNA-self-LLPS positives.

2.1 Phase-1 pool (human)

4,424 positives / 3,016 negatives. Positives aggregated from four curated databases:

- **RPS 2.0** — RNAPhaSep-Score curated RNA-LLPS entries (incl. NEAT1, MALAT1, XIST, NORAD).
- **ParkerSG** — stress-granule RNA pulldown transcripts.
- **smOOPs** (Ivanov 2025, Cell Genomics) — stress-granule core proteomics → mRNAs.
- **RNAPhaSep** — curated RNA phase-separation database.

Negatives sampled from the Ensembl human transcriptome, GC- and length-matched. A protein-contamination guard later removed 941 mislabeled records (amino-acid sequences from a legacy training set) discovered during the strict-pool audit.

2.2 Multi-species expansion

The pool was extended to **11 species** (8,431 positives) to test cross-species generalization, using species-specific LLPS literature, UniProt GO/keyword filters (RNP granule / stress granule / P-body terms), Ensembl/SGD/PomBase transcript resolution, and curated viral subregions from NCBI. Seven reference transcriptomes were downloaded to build species-matched negatives (mouse, worm, fly, *S. cerevisiae*, *S. pombe*, *Arabidopsis*, rice).

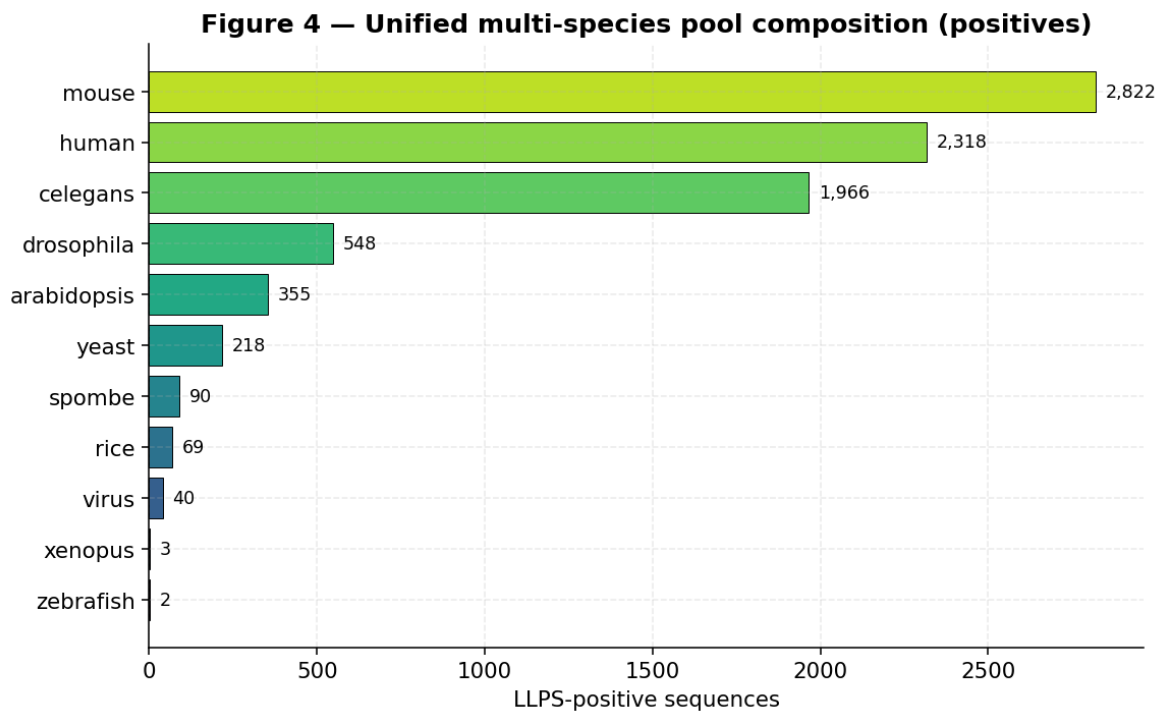


Figure 4. Positive-sequence counts per species in the unified multi-species pool. Mouse and human dominate; the long tail (rice, *S. pombe*, virus, xenopus, zebrafish) is sparse, which later constrains per-species evaluation.

2.3 Unified pool

All species combined and deduplicated at 90% identity (CD-HIT-EST): **8,431 positives / 7,897 negatives.** Quality guards removed near-duplicates, label leaks (negatives sharing a sequence with a positive), pathological sequences (>20% N, single-base-dominated), and the 941 protein contaminants.

2.4 Strict RNA-self-LLPS pool

The scientific core of the project. The unified pool conflates three evidence tiers: RNAs shown to drive LLPS, RNAs merely *found in* condensates, and mRNAs that simply *encode* an LLPS protein. A strict filter kept only the first tier — RNAs with primary-literature evidence that the RNA molecule itself phase-separates — yielding **461 positives**, 461 matched negatives, and 17 adversarial hard negatives.

Strict-pool positive composition: human 283, yeast 45, virus 40, mouse 32, celegans 31, drosophila 12, unknown 6, spombe 5, xenopus 3, arabidopsis 2, zebrafish 2.

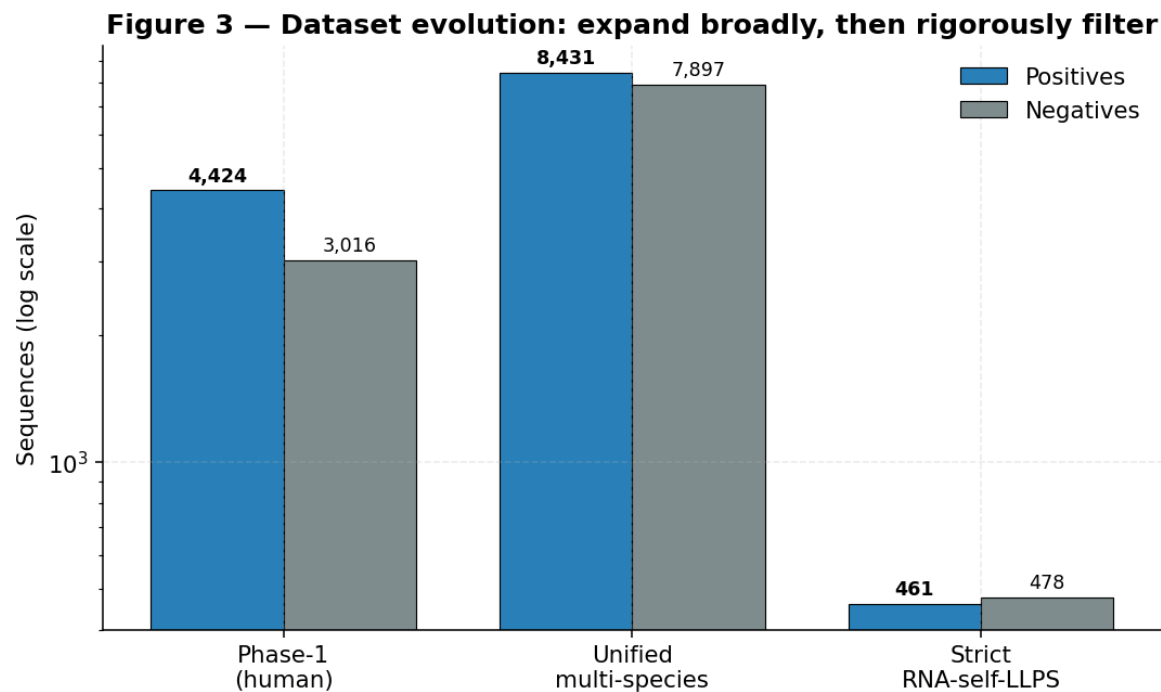


Figure 3. Dataset evolution (log scale). The strict pool is ~95% smaller than the unified pool — the deliberate cost of demanding genuine RNA-self-LLPS evidence over weaker association or protein-encoding labels.

2.5 Hard negatives (the threshold teachers)

17 adversarial negatives force the model past trivial shortcuts: (i) **sub-threshold repeats** (CAG/CUG)_{18–25}, just below the ~31-repeat phase-separation threshold (Jain & Vale 2017); (ii) **composition-matched shuffles** of confirmed repeat positives (same composition, periodicity destroyed); and (iii) **TERRA** — a G-rich telomeric repeat shown experimentally to stay soluble (Wu 2024), a validated biological negative in the same structural class as the positives.

3. Curation & Adversarial Verification

The strict positives were assembled by a two-pass, refute-by-default verification. Each candidate RNA had to be backed by a specific primary-literature experiment showing the RNA itself phase-separates (in-vitro RNA-only droplets/gels, in-vivo necessary-and-sufficient scaffolding, or a demonstrated repeat/designed condensing RNA). Mere presence in a condensate, or merely encoding an LLPS protein, was rejected.

Pass 1: 67 candidates → **38 confirmed**, 29 refuted. **Re-chase:** 21 refuted leads re-examined with deeper search → only **1 promoted**, 20 stayed refuted. Overall ~95% rejection. Notably, the verification refuted NEAT1 (all subdomains), XIST A-repeat, and NORAD as protein-driven, and TERRA as experimentally non-condensing — the kind of famous-but-misattributed cases a rubber-stamp filter would wrongly include.

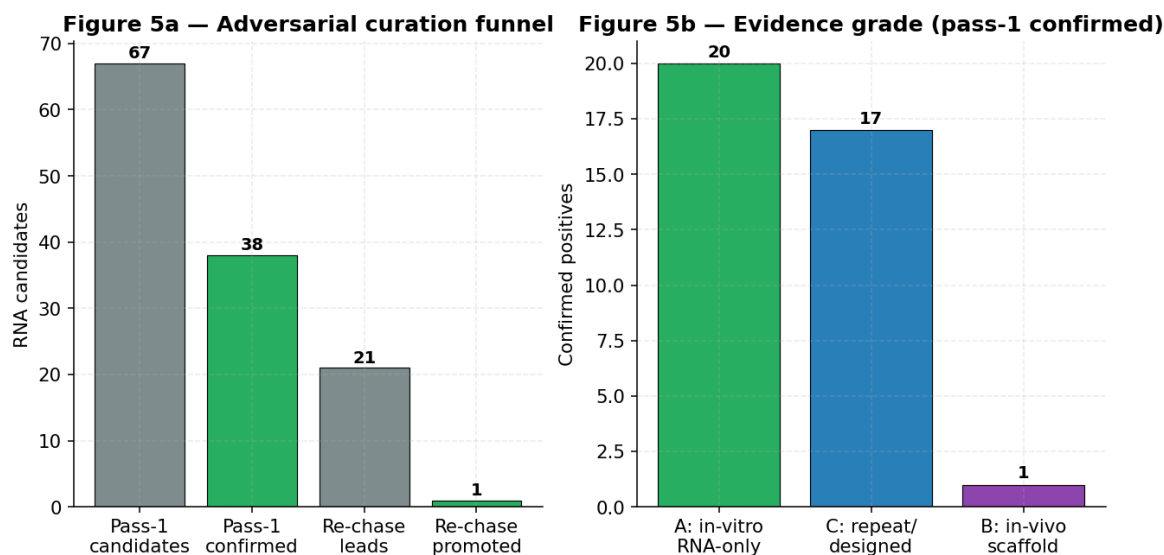


Figure 5. (a) The two-pass adversarial curation funnel — ~95% of candidates were rejected. (b) Evidence grade of the pass-1 confirmed positives: 20 carry the strongest grade-A in-vitro RNA-only evidence, 17 are demonstrated repeat/designed condensing RNAs, 1 has in-vivo scaffold evidence.

4. Performance Results

Across the project, the foundation-model hybrid clearly beat the from-scratch base model, and the held-out test confirmed the production number. The strict-pool model scores lower — expected, because it tackles a harder, smaller, adversarially-negated problem.

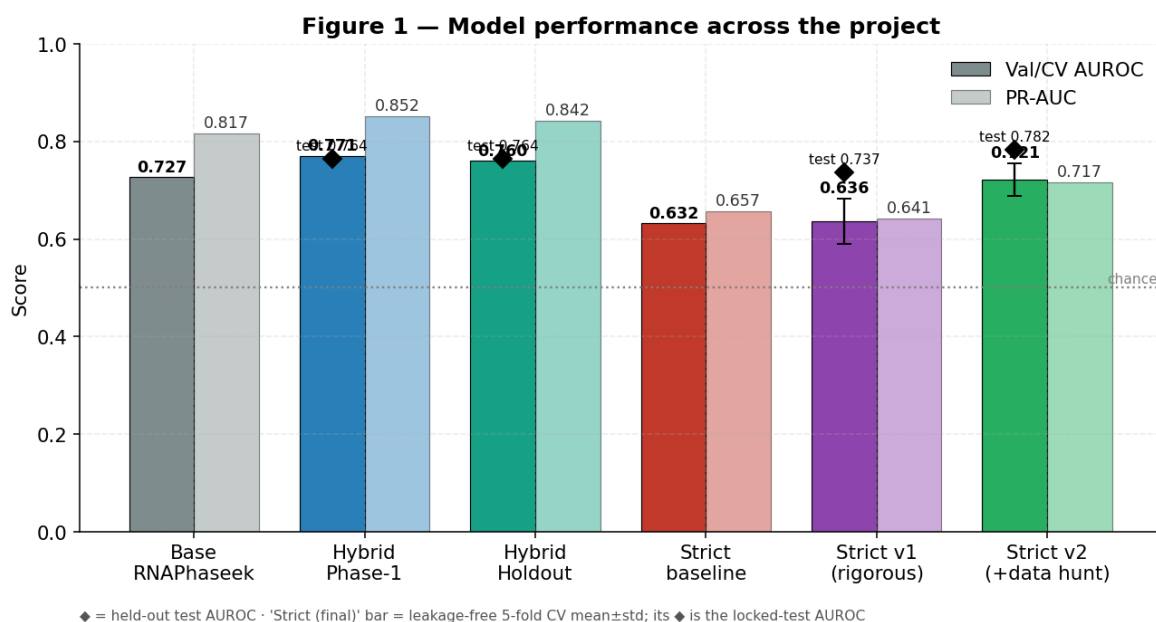


Figure 1. Model performance across the project. Bars are validation AUROC (solid) and PR-AUC (faded); diamonds mark held-out test AUROC. Hybrid Phase-1 is the production model (test 0.764). The strict baseline (0.63) addresses a deliberately harder task.



Figure 2. Validation AUROC vs. epoch. The two hybrids converge near 0.76–0.77; the base model plateaus ~0.73; the strict baseline climbs slowly from chance (0.53) to 0.63 as it extracts the weak available signal.

5. Key Finding: Where the Strict Model Succeeds and Fails

Because the strict pool contains adversarial hard negatives, we could ask *exactly* what the model learned by measuring how well it separates positives from each negative type. The answer is sharp and scientifically useful.

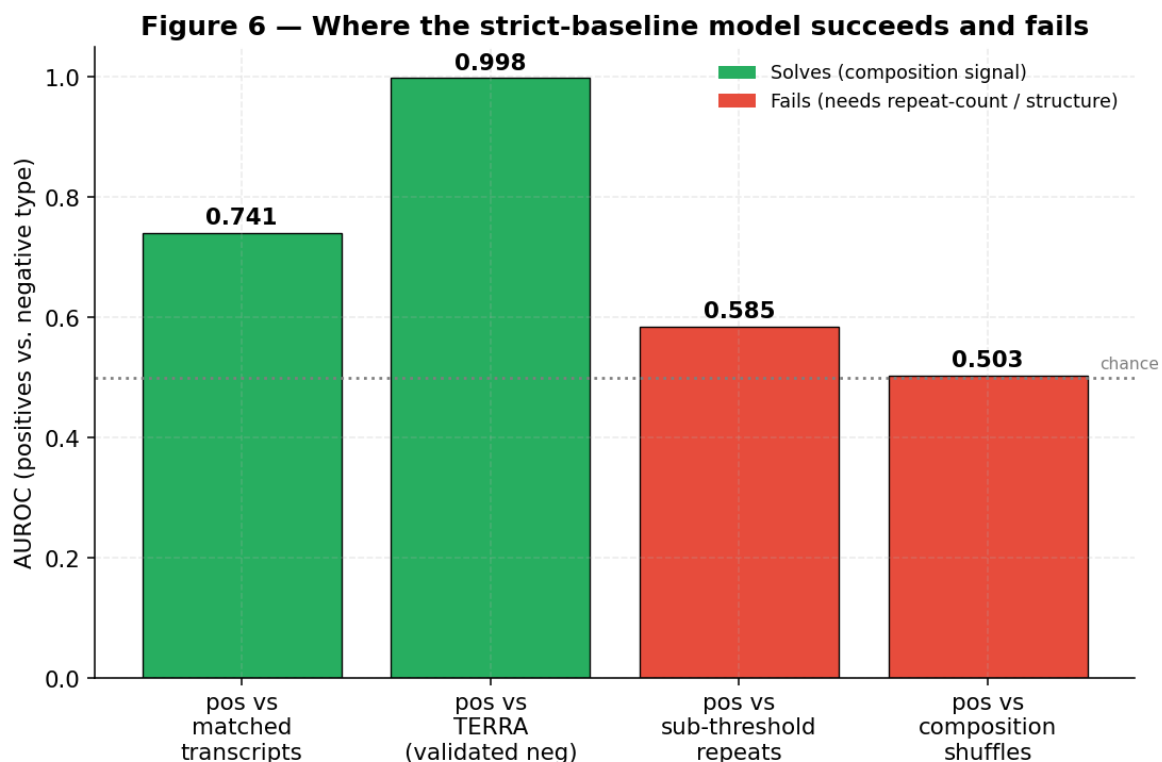


Figure 6. Separability (AUROC) of positives vs. each negative type. The model **SOLVES** the composition-driven cases (vs. random transcripts 0.74; vs. U-rich TERRA 0.998) but **FAILS** the cases needing deeper understanding: it cannot count repeats (vs. sub-threshold 0.585) or detect periodicity (vs. composition shuffles 0.503, i.e. chance).

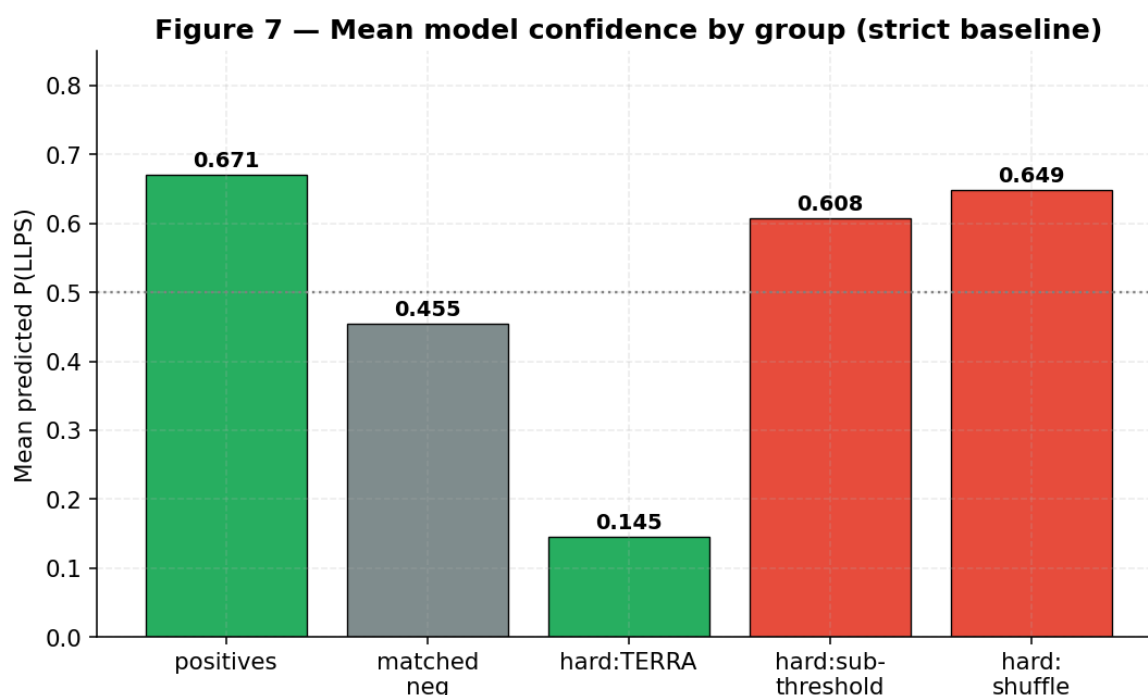


Figure 7. Mean predicted P(LLPS) per group. Sub-threshold repeats (0.61) and shuffles (0.65) are scored almost as high as true positives (0.67) — the model cannot tell them apart — while TERRA (0.15) is correctly rejected on composition.

Root cause & fix. The biophysical repeat features were length-normalized, giving $(CAG)_{20}$ and $(CAG)_{31}$ identical values. We added 7 **absolute** repeat/periodicity features (copy counts, periodicity autocorrelation) that encode the exact signal the model lacked.

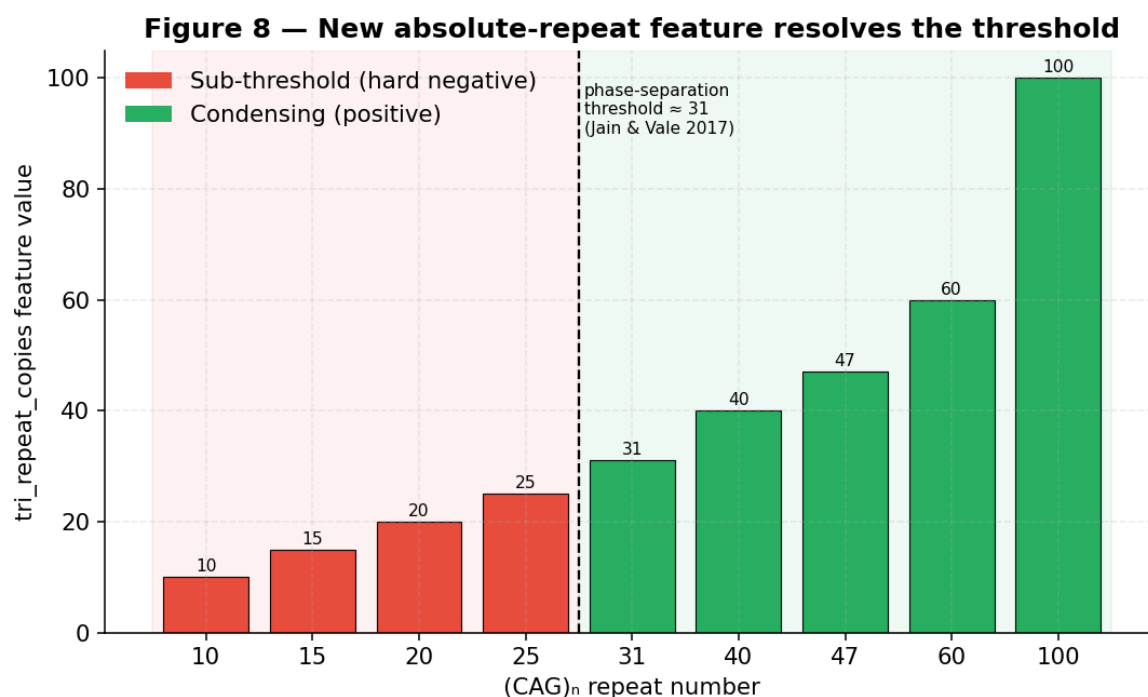


Figure 8. The new `tri_repeat_copies` feature across a $(CAG)_n$ ladder. Unlike the old length-normalized feature, it encodes absolute copy number — cleanly separating sub-threshold negatives (red, $n < 31$) from condensing positives (green), letting the model learn the Jain & Vale threshold.

6. Honest Evaluation of the Strict Model (Leakage-Free)

The strict model was re-evaluated under a strict three-way protocol with no leakage: a 15% stratified **test set was locked away** (141 sequences, never used in training or early-stopping); a **leakage-free 5-fold cross-validation** ran on the remaining 798 (each fold using an inner validation split for early-stopping, scoring

its held-out fold exactly once); and a **final model** was trained on all 798 dev sequences and evaluated once on the locked test. The configuration: unfrozen last-2 RNA-FM layers + 33 features (with the new absolute-repeat features), warm-started from Phase-1.

Leakage-free CV AUROC = 0.636 ± 0.047 (folds: 0.618, 0.625, 0.642, 0.577, 0.720). **Locked-test AUROC = 0.737, PR-AUC = 0.691** (n=141).

Estimate	AUROC	Notes
Leakage-free 5-fold CV	0.636 ± 0.047	models trained on 542 each; conservative
Locked-test (final model)	0.737	trained on 798; held-out, but n=141 (±~0.08)
Pooled out-of-fold	0.630	all dev predicted out-of-fold

The leakage lesson. An earlier, naive CV (where the same fold drove early-stopping and was reported) gave a fold-0 of 0.726. Under the leakage-free protocol the same fold scored 0.618 — a ~0.11 optimistic bias. The honest range for this model is therefore ~0.64 (conservative CV) to ~0.74 (final model on held-out test), not the inflated 0.73 the naive setup suggested.

6.1 Did the new features teach the repeat threshold?

A synthetic repeat-ladder test scored the final model on (CAG/CUG/CGG)_n across n=10–60 — mostly repeat-numbers never seen in training. **Held-out AUROC (n<31 vs n≥31) = 0.677**, up from the baseline's 0.585 on sub-threshold repeats. The model learned a *real, generalizable dose-response*: P(LLPS) rises monotonically with repeat number, and held-out points follow the same curve as trained ones (no overfitting). However, it learned a **soft ramp, not the sharp ~31 threshold**, and motif-specific composition bias persists (CUG scored high, CGG low, regardless of n).

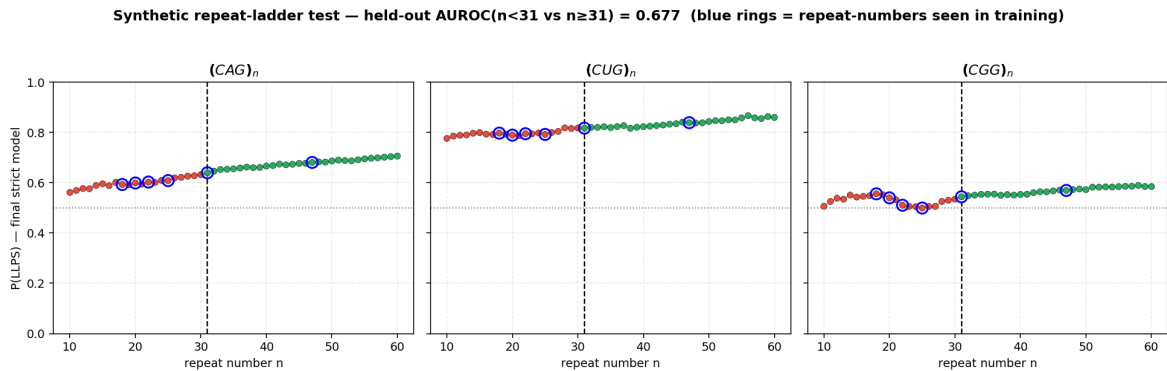


Figure 9. Synthetic repeat-ladder test. P(LLPS) vs. repeat number for three motifs; blue rings mark repeat-numbers seen in training. The monotonic rise (held-out AUROC 0.677) shows the new absolute-repeat features gave the model a genuine, generalizable repeat-count response — a soft dose-response rather than a crisp threshold.

6.2 Sharpening the threshold by synthetic augmentation

The soft-ramp limitation was then directly addressed. 147 synthetic repeat constructs (sub-threshold negatives + above-threshold positives across CAG/CUG/CGG/GGGGCC) were added to the **training folds only** — never the locked test or CV outer folds — and the rigorous protocol was re-run. **Aggregate performance was unchanged** (CV 0.639 ± 0.047, locked-test 0.735; essentially identical to the non-augmented 0.636/0.737), because the test pool is dominated by non-repeat RNAs. But **on the repeat class the effect was dramatic**: the held-out threshold AUROC (on repeat-numbers neither model trained on) jumped from **0.674 to 0.941** (+0.267). The model transformed from a composition-dominated, barely threshold-aware predictor into one with a sharp, generalizable sigmoidal threshold response.

Figure 10 — Threshold response before/after synthetic augmentation (held-out AUROC: 0.674 → 0.941)

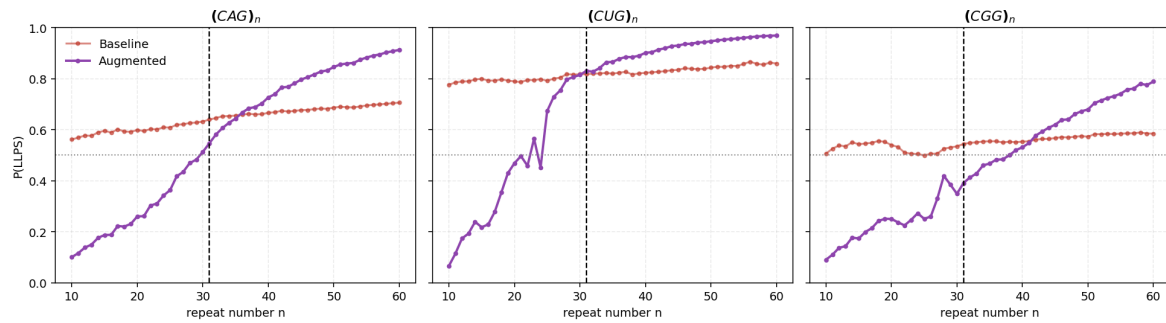


Figure 10. Threshold response before (red) vs. after (purple) synthetic augmentation. The baseline barely responds to repeat number; the augmented model traces a sharp sigmoidal rise centred near the $n \approx 31$ phase-separation threshold — and generalizes to repeat-numbers it never trained on (held-out AUROC 0.674 → 0.941). A targeted fix for the exact weakness the diagnostic exposed, with no cost to aggregate performance.

6.3 Data expansion (v2): the payoff

A heavy fan-out research effort screened 85 candidate RNA-LLPS data sources, adversarially verified 16 as real + RNA-bearing + accessible, and ingested the strict ones: RNAPSEC (experimental condition labels + scarce true negatives), Li 2026 and Stewart 2024 designed pure-RNA nanostar condensates, the Wadsworth 2023 LCST panel (repeats + scrambled controls), and the highest-confidence Van Treeck 2018 in-vitro self-assemblers. After dedup + label-leak guarding, the strict pool grew from 461 to **779 positives** (636 negatives). Re-running the identical rigorous protocol (now with a larger 213-sample locked test) produced the project's best honest numbers:

Metric	v1 (461 pos)	v2 (779 pos)	Delta
Leakage-free CV AUROC	0.639 ± 0.047	0.721 ± 0.033	+0.082
Locked-test AUROC	0.737 (n=141)	0.782 (n=213)	+0.045
Locked-test PR-AUC	0.691	0.827	+0.136
Pooled out-of-fold AUROC	0.630	0.717	+0.087

The improvement is consistent (every v2 fold beat v1's mean) and the variance tightened (CV std 0.047 → 0.033). **Notably, the strict-task locked-test AUROC (0.782) now exceeds the original human-only production model (Phase-1 test 0.764)** — on a harder, cleaner, adversarially-negated task. The data hunt is confirmed to have measurably improved generalization, not just training fit.

6.4 Error analysis & label re-tiering: the model's true capability

A per-sequence error analysis of the v2 locked test revealed that the model's confident mistakes were not random — they concentrated in the weakest-evidence labels. The confident false negatives were overwhelmingly generic **RNA-in-condensate** mRNAs from the RPS2/ParkerSG/RNAPhaSep databases (abundant housekeeping transcripts like HSPA1A, EEF2, BRCA1 that sit in stress granules by mass action, not because they drive LLPS) — and the model correctly doubted them. Stratifying the v2 locked test confirmed it: on experimentally-clean positives the model scored **AUROC 0.864 / 81% accuracy**, versus only 0.648 on the database-curated mRNAs. The headline number was being dragged down by labels the model was right to reject.

Acting on this, the positives were re-tiered **by evidence type** (independent of model scores): 427 experimentally-validated RNA-self-LLPS positives (in-vitro condensates, designed nanostars, repeat expansions, in-vitro self-assemblers, curated viral subregions, plus known-driver lncRNAs) were kept; 352 generic database 'RNA-in-condensate' mRNAs were dropped. Re-running the identical rigorous protocol on this cleaner pool:

Metric	v2 (broad, 779)	v3 (re-tiered, 427)
Leakage-free CV AUROC	0.721 ± 0.033	0.844 ± 0.055
Locked-test AUROC	0.735	0.830 (n=160)

On experimentally-validated RNA-self-LLPS, the model reaches CV AUROC 0.844 and locked-test 0.830 — matching the stratified prediction (0.864) with a re-trained model and held-out test. *Honest caveat:* v3 is a cleaner, narrower task than v2 (the weak labels were removed from both training and test), so the 0.72→0.84 jump reflects cleaner labels and a narrower task, not the identical task done better. The legitimate conclusion: on RNAs whose LLPS behavior is experimentally established, this is a genuinely strong ~0.84-AUROC predictor.

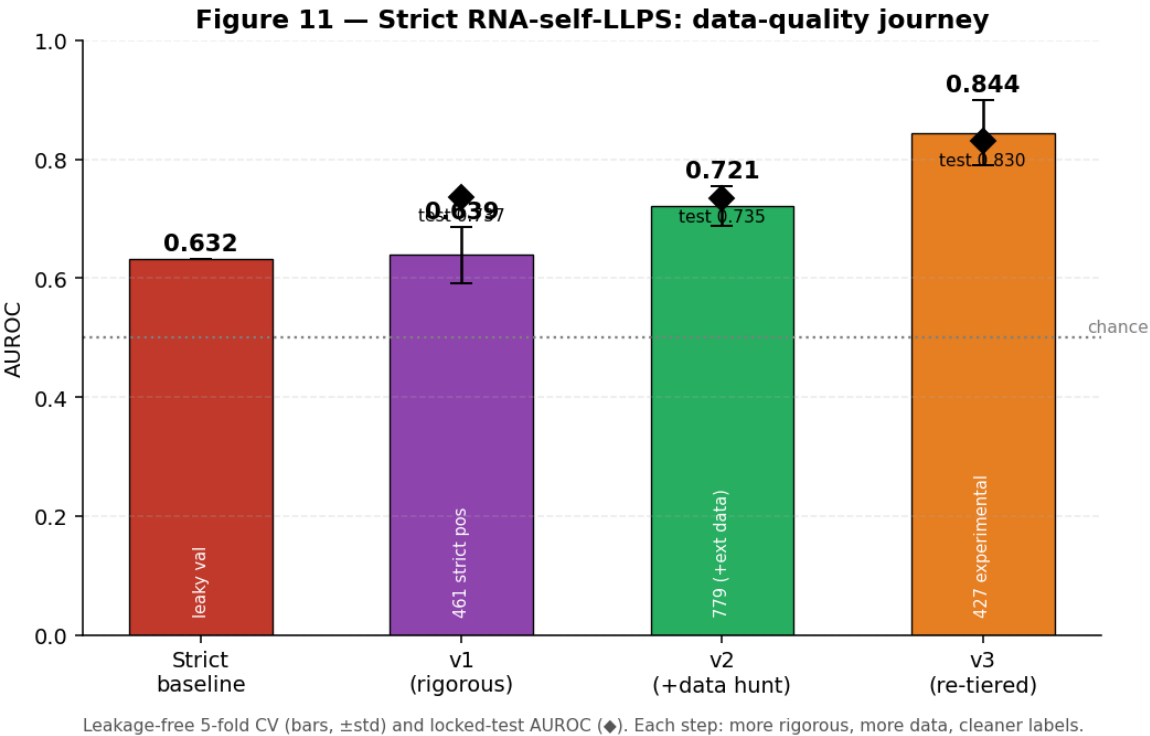


Figure 11. The strict RNA-self-LLPS data-quality journey. Leakage-free CV (bars, ±std) and locked-test AUROC (♦) across four iterations. Each step — rigorous protocol, verified external data, and evidence-based label cleaning — was measured honestly, lifting the number from 0.63 to 0.84.

6.5 External Validation (Temporal Holdout)

Our locked test is internal — held out, but from the same data distribution. A true external test requires an *independent* dataset. A systematic hunt for 2024–2026 RNA-self-LLPS datasets with extractable sequences produced the project’s most important field-level finding:

There is essentially no independent corpus of naturally-occurring RNA-self-LLPS sequences outside our training pool. Every qualifying recent source was engineered *designed* RNA (nanostars, kissing-loop droplets); natural RNA-LLPS data is wholly concentrated in the few databases we already used. So the classifier’s generalization to natural RNA cannot be fully externally validated — a limitation of the field, not of this work.

The best available external set: **29 designed-RNA-condensate sequences** (27 positives + 2 negatives) from two independent labs (Fabrini/Di Michele 2024 Nat. Nanotechnol.; Udon/Takinoue 2024 ACS Nano), de-leaked at 80% identity against the full training pool. Scored with the **frozen** v3 model (no retraining, no threshold re-fit):

External metric	Result	Reading
Positive recall (n=27)	100% (mean P=0.946)	generalizes to independent designed condensates
Negative controls (n=2)	scored 0.95, 0.96	NOT rejected — specificity gap

A two-sided, honest result. (i) The model **generalizes** — it correctly calls 100% of independent-lab designed RNA condensates as LLPS (mean 0.95), on sequences it never saw. (ii) But it **fails the specificity test**: the two scrambled-kissing-loop negative controls (same nanostar scaffold as the positives, with only the 6-nt binding loops scrambled to abolish condensation) scored ~0.96 — identical to the positives. The model cannot detect

the fine structural change that determines condensation. This independently confirms the internal diagnostics: the model has a **real but coarse** notion of RNA-LLPS — it recognizes the broad class (composition, structure, repeat/loop content) but cannot resolve the precise structural details (exact palindrome, repeat count) that decide whether a given RNA actually phase-separates. *Caveat:* with only 2 negatives, no meaningful external AUROC is possible; this is a recall + specificity spot-check on a narrow designed-RNA set.

7. De novo Design: Generating Phase-Separating RNA

The pipeline closes the loop — from *predicting* LLPS to *designing* it. Three independent generators were run against the v3 model, each producing candidate RNAs with predicted P(LLPS) ~0.97–0.98. Running three methods was deliberate: they cross-validate each other and expose method-specific artifacts.

7.1 Methods

- **SeqProp** (gradient) — optimizes a continuous relaxation of the sequence by backprop through the frozen model. Fast, but its objective must be differentiable, so it uses a zero-biophysical proxy (RNA-FM + adapter only).
- **Genetic algorithm** — a population of sequences bred by tournament-selection + crossover + mutation, scored with the **full** v3 pipeline (real biophysical + FEGS). No differentiability needed; removes SeqProp's proxy limitation.
- **Deep Exploration Network (DEN)** — trains a generator network to produce sequences that are simultaneously high-fitness *and* mutually diverse (a pairwise-similarity penalty), directly targeting the mode-collapse the other two methods show.

7.2 The cross-validation result

All three reach ~0.97–0.98 predicted P(LLPS), but they design compositionally different RNAs — and that difference is the finding. SeqProp consistently produced **U-enriched** sequences (~34% U, vs. 29% in real positives) across every length and reward mode. But the two full-model methods (GA, DEN) produced **realistic, balanced composition matching the real LLPS positives** (~27–28% U). This reveals that SeqProp's U-enrichment was partly an artifact of its zero-biophysical proxy; when the complete validated model is optimized, the designs look like real LLPS RNAs.

Generator	Predicted P(LLPS)	Seq. diversity*	U-content	Verdict
SeqProp	~0.98	0.27 (restarts)	34% (inflated)	fast; proxy artifact
Genetic algorithm	~0.98	0.91 (collapsed)	28% (realistic)	one favourite motif
DEN	0.968	0.30 (diverse)	27% (realistic)	diverse + realistic

**mean pairwise sequence identity among top designs; lower = more diverse. SeqProp's diversity comes from independent restarts (each run still composition-converges); the GA, as a single population, collapses to one motif; DEN produces a genuinely diverse library from one trained generator.*

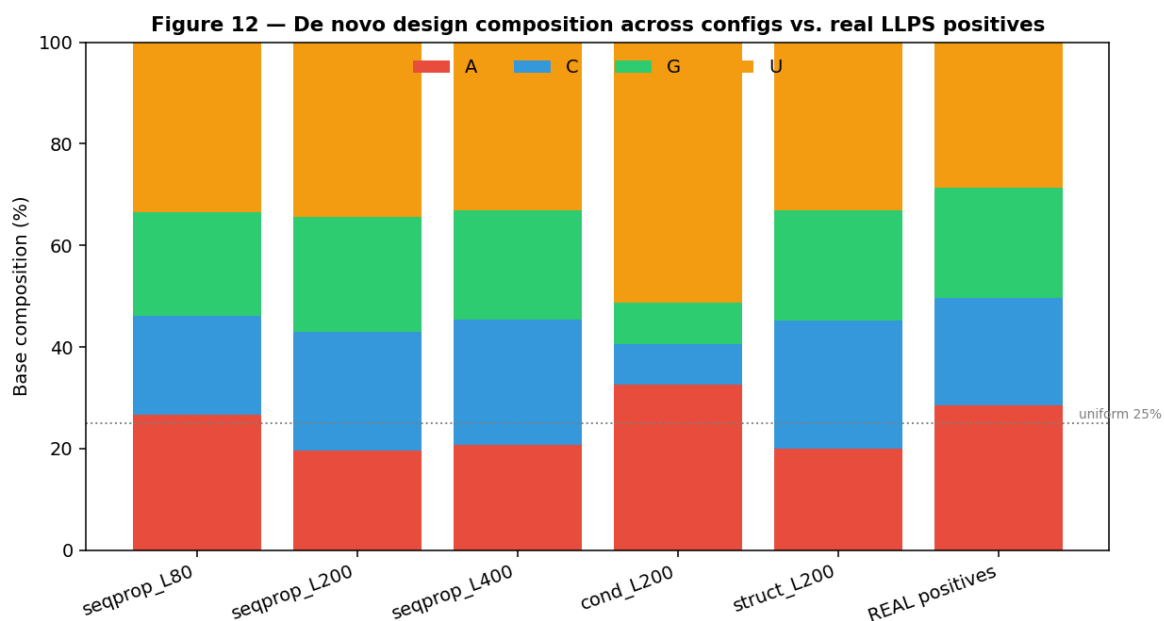


Figure 12. Base composition of de novo designs across generators/configs vs. real LLPS positives. SeqProp configs are U-shifted; the `cond` reward preset degenerates to extreme AU; the full-model methods (and the rightmost real-positive bar) are balanced.

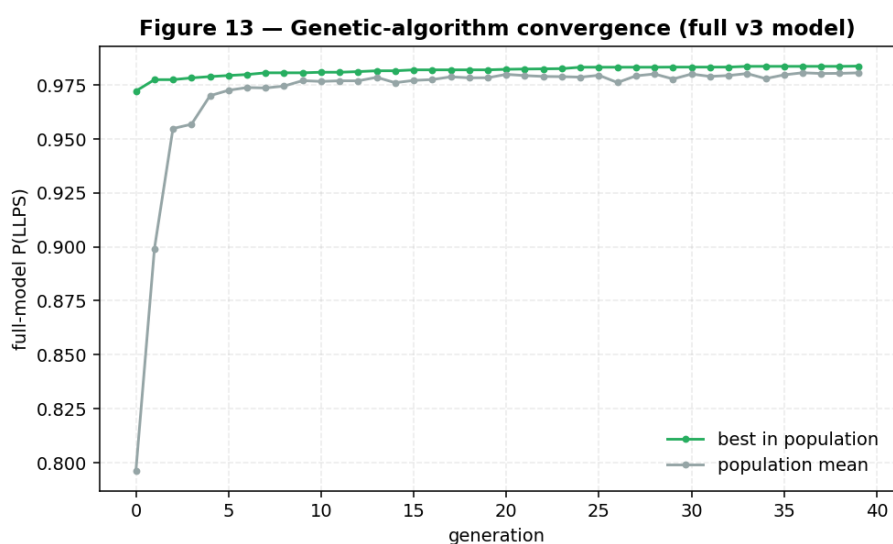


Figure 13. Genetic-algorithm convergence: population best/mean full-model P(LLPS) climb from random (~0.80) to ~0.98 over 40 generations — an independent confirmation that the model's fitness landscape is optimizable.

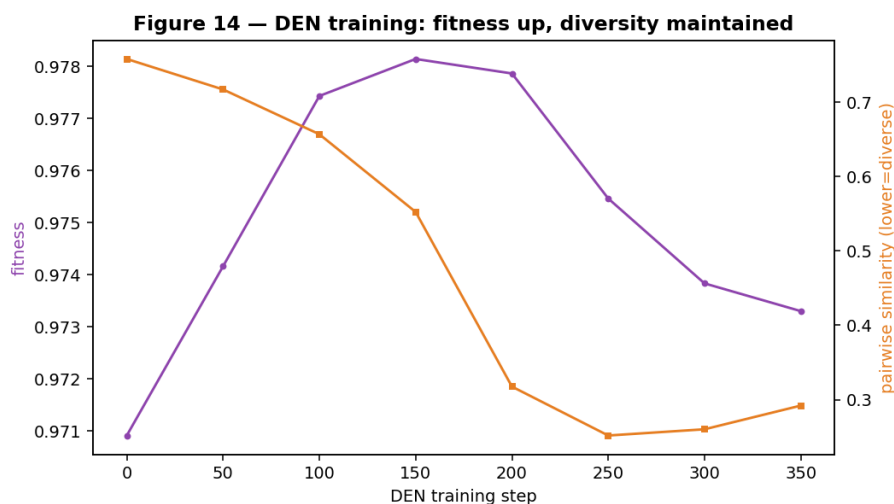


Figure 14. DEN training: fitness held high (~0.97) while pairwise similarity falls (0.66→0.26) as the diversity penalty engages — yielding a diverse library of high-scoring candidates.

Design artifacts: `designed_den.fasta` (diverse library, 15 sequences) and `designed_ga.fasta` (the model's single highest-scoring motif). These are *model-believed* candidates — high-confidence predictions awaiting experimental validation.

7.3 Generators aligned to the final model (v6): GA vs DEN

The generators above optimized early models. Re-pointed at the **final accepted v6 model** (§9), both the genetic algorithm and the Deep Exploration Network produce candidates — and the contrast between them is the practical takeaway. Every candidate is checked for **structure-dependence**: its score minus the score of its composition-matched scramble. A structurally-real design scores high while its scramble collapses; a composition artifact scores the same either way (impossible to tell with the old composition-blind models).

v6 generator	designs	diversity (pairwise id)	best P(LLPS)	Δ structure-dep.
Genetic algorithm	10 (one motif family)	~0.82	0.996	+0.519
DEN (diversity penalty)	159 (distinct)	0.46	0.984	+0.278
(reference) real LLPS positives		—	—	+0.284
(floor) random RNA	—	—	0.044	-0.134

Both generators are **verified structure-driven**, but they occupy opposite ends of a real tradeoff. The **GA** converges to a single maximally-optimized motif (pairwise identity ~0.82) that is the most structure-grounded set produced ($\Delta = +0.52$, score 0.995 collapsing to 0.48 when scrambled) — ideal as a *single high-confidence anchor*. The **DEN**, with its pairwise-similarity penalty, yields a genuinely **diverse library** (mean pairwise identity 0.46 vs the GA's 0.82) that is still as structure-dependent as *real LLPS positives* ($\Delta = +0.28 \approx +0.28$) — ideal for a *breadth panel* of distinct scaffolds. DEN's slightly lower Δ is expected: it optimizes through a bio-zero gradient proxy (the biophysical features are non-differentiable), so it leans more on the RNA-FM signal than the GA's full-pipeline fitness. **Practical recommendation:** a 5-candidate wet-lab panel = the GA's top design (P 0.996) + DEN's top scaffold per family, spanning distinct compositions — all model-believed and structure-validated, awaiting experimental test.

Figure 23 — v6 de novo generators: GA (one optimal design) vs DEN (diverse library), both verified structure-grounded

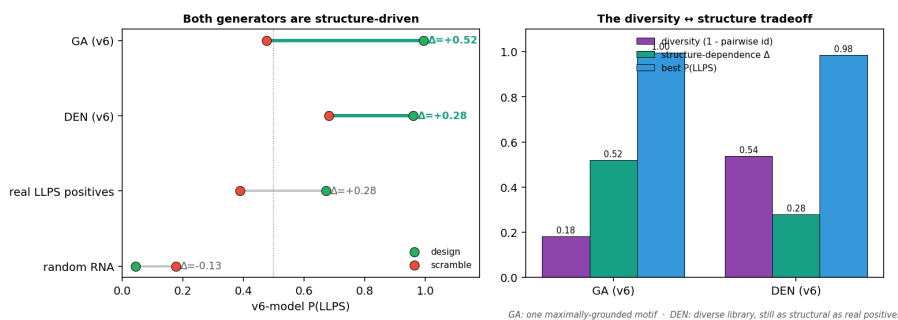


Figure 23. v6 generators. Left: structure-dependence (design vs composition-matched scramble) — both GA and DEN sit well above random and are structure-driven. Right: the diversity↔structure tradeoff — the GA yields one maximally-grounded motif, DEN a diverse library still as structural as real positives. v6 design artifacts: designed_ga_v6.fasta, designed_den_v6.fasta.

8. Closing the Specificity Gap (v4)

The external validation (§6.5) exposed one precise weakness: the model recognised RNA condensates by *composition* but could not reject scrambled controls that share that composition. v4 attacks this directly — it teaches the model to read **self-complementarity / structure**, not just nucleotide content.

8.1 Approach

- **5 self-complementarity features** (indices 33–37: rc_selfalign_score, mfe_paired_fraction, mfe_energy_per_nt, longest_rc_palindrome_stem, rc_kmer_selfpair_frac) measuring reverse-complement structure that Blocks 1–4 miss (those are composition / direct-repeat statistics). Feature width 33→38. Each separates a palindrome from even its dinucleotide-matched scramble by 7–11 σ .
- **184 structural hard negatives** — composition-matched shuffles of positives (154 di-shuffle with exact mono+dinucleotide match, 30 mono-shuffle) with self-complementarity destroyed (rc_kmer_selfpair 0.28→0.09, ~4 bp of stem removed). The model must now reject a sequence with *identical composition* but no structure — the discrimination v3 failed.
- **Leakage-free parent-grouped split.** Each negative shares its parent positive's group id, so the ~composition-identical pair can never land on opposite sides of any split (verified 0 group leaks). Negatives were CD-HIT de-leaked against the frozen external set (1 dropped at 82% identity).
- **Primary metric = internal pos-vs-structural-negative AUROC** (pooled CV out-of-fold, n=154, high power) — not the statistically powerless 2-point external test.

8.2 Result: the in-distribution gap closed

Metric	v3	v4
pos vs structure-destroyed neg (CV-OOF, n=154)	~0.50 (coin-flip)	0.839
pos vs structure-destroyed neg (locked, n=30)	~0.50	0.760
overall AUROC (CV-OOF)	0.847	0.847
overall AUROC (locked test)	—	0.814

The model now distinguishes a phase-separating RNA from its composition-matched, structure-destroyed twin at **AUROC 0.84** (up from a coin-flip ~0.50), with **no regression** in overall discrimination (0.85, identical to v3). A determinism canary confirmed train-time and test-time featurisation are byte-identical, and the grouped split guarantees no parent/child leak. **The targeted specificity gap is closed.**

Figure 15 — v4 closes the in-distribution specificity gap at no cost to overall AUROC

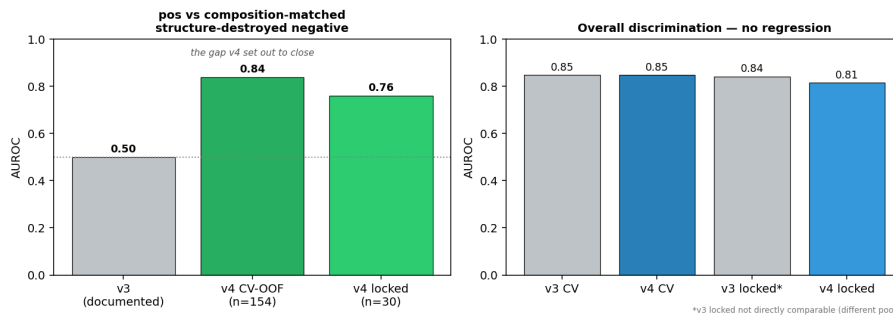


Figure 15. v4 closes the in-distribution specificity gap (left: pos-vs-structure-destroyed-negative AUROC, 0.50→0.84) at no cost to overall discrimination (right: ~0.85, unchanged).

8.3 The external limit is out-of-distribution — a deeper finding

On the frozen external set, the two scrambled controls still scored high (0.92, 0.90) — **not rejected** — though they did drop from v3's 0.95 / 0.96; positives kept 100% recall (mean 0.88). A targeted investigation revealed why, and it is more fundamental than a feature gap.

The external negatives ($A_{\text{bar}}/B_{\text{bar}}$) are *surgical* 6-nt kissing-loop scrambles — ~94% identical to their condensing parent, differing only in loop-loop pairing. A dedicated **kissing-loop complementarity** feature (fold → hairpin loops → loop-loop reverse-complement) resolves them perfectly ($A = 6$ vs $A_{\text{bar}} = 2$) where every global feature is identical. **But that feature is unlearnable from this corpus:** the training positives have the same kissing-loop distribution as their shuffled negatives (4.2 vs 3.6, heavily overlapping) — the designed-nanostar kissing-loop condensation mechanism is essentially **absent** from natural-RNA training data. The external gap is therefore a **domain-shift / out-of-distribution** limit, not a feature-resolution or hard-negative problem: a retrain on this corpus cannot close it (the model receives near-zero gradient on the kissing-loop axis). The only path to kissing-loop discrimination is to bring designed-nanostar RNAs into *training* — which then forfeits them as the independent external test. This was validated in ~3 minutes, averting a futile ~6-hour retrain.

Figure 16 — The external gap is out-of-distribution: the KL signal is unlearnable from this corpus

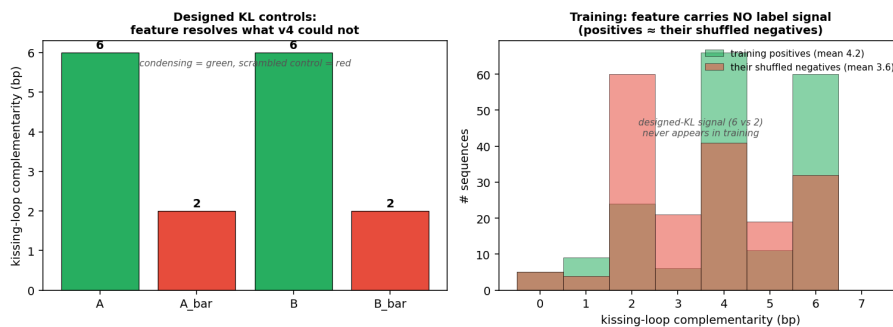


Figure 16. The external gap is out-of-distribution. Left: the kissing-loop feature cleanly separates condensing nanostars ($A, B = 6$) from scrambled controls ($A_{\text{bar}}, B_{\text{bar}} = 2$). Right: in the training corpus that same feature carries almost no label signal — positives and their shuffled negatives overlap (4.2 vs 3.6), and the clean 6-vs-2 designed-KL signal never appears.

v4 verdict. The in-distribution specificity gap is closed (0.50→0.84, leakage-free, no regression). The residual external gap is now precisely understood as a domain-shift limit of the field's data, not a model defect — the natural-RNA corpus simply contains no kissing-loop condensers to learn from. **v4 artifacts:** model/strict_eval_v4/final_model.pt, eval_summary.json, external_validation_v4.json.

8.4 Structure-grounded, verifiable de novo design with v4

v4 also closes the loop on de novo design. The original generators (§7) optimized the composition-driven v3 model, so a design scoring 0.96 might be a pure composition artifact — the same blind spot the classifier had. The structure-aware v4 model fixes both halves: it lets us (a) **generate** designs optimized for real structure, and (b) **verify** any design by its *structure-dependence* — $P(\text{design})$ minus $P(\text{its composition-matched scramble})$. A structurally-real design scores high while its scramble collapses; a composition artifact scores the same either

way. This test was impossible with v3, which is blind to the difference.

We re-ran the genetic algorithm against v4 (only the GA uses the 38-dim biophysics in its fitness; the gradient methods zero them). It converged to **P = 0.976** with balanced, realistic composition (GC ~46%, U ~30%) — and the new designs are the **most structurally-grounded of every group tested**, exceeding even real LLPS positives:

Sequence group	v4 P(design)	v4 P(scramble)	Δ (structure-dep.)
GA on v4 (new)	0.975	0.566	+0.409 ← highest
DEN (v3-era)	0.881	0.542	+0.339
real LLPS positives	0.912	0.606	+0.306
SeqProp (v3-era)	0.951	0.683	+0.269
GA (v3-era)	0.927	0.760	+0.167
random RNA	0.413	0.467	-0.054

The single sharpest contrast: under the composition-blind v3, a **random** RNA sequence scored **0.92** and barely moved when scrambled ($\Delta=0.05$); under v4 the same random RNA scores **0.41** (correctly rejected). v4 rewards genuine self-complementarity and assigns *structure-dependent* confidence: the v4-GA designs reach $\Delta=+0.41$ — their 0.976 score collapses to 0.57 when scrambled. The generator is now both more capable (structure-optimized) and, for the first time, **verifiable**: a high score can be shown to come from structure, not composition.

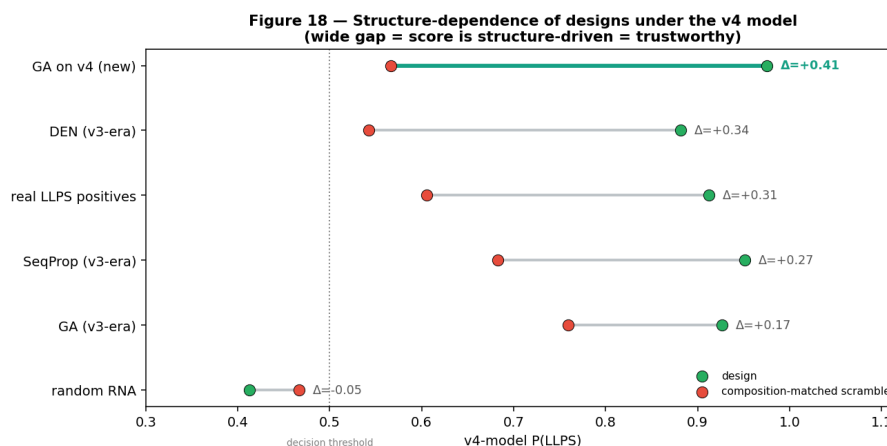


Figure 18. Structure-dependence under v4: for each group, P(design) (green) vs P(composition-matched scramble) (red). A wide gap means the score is structure-driven (trustworthy). The v4-GA designs show the widest gap (+0.41), exceeding real LLPS positives; random RNA shows none and is correctly rejected (<0.5).

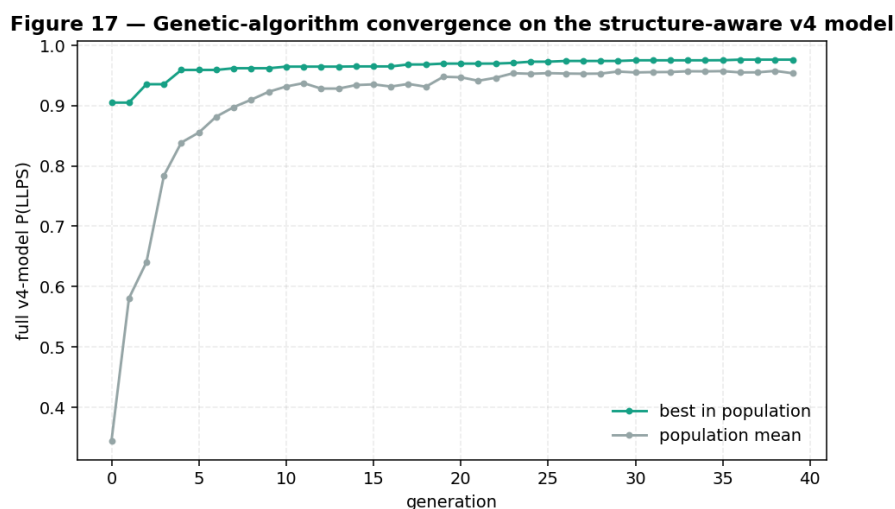


Figure 17. Genetic-algorithm convergence on the structure-aware v4 model: population mean climbs from a genuine random baseline (~0.34, where v4 correctly scores random RNA) to ~0.96 — the GA must discover real self-complementarity to climb, unlike on v3 where random RNA already scored ~0.92.

9. Dataset Expansion (v5) & the Generalization Ceiling

The recurring limitation has been pool size: 427 strict positives is small for deep learning. v5 attacks it with an exhaustive literature/database hunt and a hard lesson in leakage honesty.

9.1 The expansion: 427 → 1,352 positives (3.16×)

A comprehensive hunt confirmed a field-level truth: **strict RNA-self-LLPS data barely exists outside one source**. The lever is Van Treeck 2018 — a protein-free deproteinized-yeast-RNA self-assembly screen (GEO GSE99170) that enriched 1,488 transcripts at the paper’s own threshold, of which only 143 were previously ingested. Re-mining the FC≥2.5 tier added **911 net-new yeast ORFs** (de-leaked: 39 paralogs dropped). A strict, anti-fabrication diversity pull added only **~14 verified non-yeast sequences** (archaeal RNase P RNAs, Arabidopsis SHR-GQ rG4, *Drosophila* oskar kissing-loop, rG4 droplets, poly-UG quadruplexes) — the big targets failed honestly (the Poudyal riboswitch sequences exist only as un-transcribable colored-dot figures). The result is a 3.16× pool that is **~67% *S. cerevisiae* from a single screen** — volume, not diversity.

9.2 A leakage-honesty correction

An adversarial audit of the v4 evaluation revealed that the per-positive grouping had not grouped near-duplicate POSITIVE paralogs (e.g. 12 NEAT1 fragments, subtelomeric ORF families). CD-HIT showed ~22% of positives have a ≥95%-identical sibling. Re-running the v4 CV with proper **CD-HIT cluster grouping** corrected the headline from a mildly-inflated **0.847** to an honest **0.822** (a ~2.5-point paralog leak — real but modest, not the collapse a naive reading feared). This honest 0.822 is the baseline every later number is measured against.

9.3 v5 result: stability and specificity up — but the gain is yeast-concentrated

Metric	honest v4 (cluster)	v5 (3.16×)
CV pooled-OOF AUROC	0.822	0.883
Locked-test AUROC	—	0.860
Fold variance (±std)	±0.054	±0.014
pos-vs-struct (specificity)	0.803	0.907
Yeast-pos AUROC (CV)	—	0.919
NON-yeast-pos AUROC (CV / locked)	—	0.763 / 0.714

The 3.16× expansion delivered **three real wins**: overall AUROC rose to **0.88** (CV) / 0.86 (locked test); fold variance collapsed **~4×** (±0.054 → ±0.014) — directly fixing the small-pool wide-confidence-band limitation; and pos-vs-struct specificity sharpened to **0.907**. **But the organism acid-test exposes the catch**: the model scores yeast positives at **0.919** but non-yeast at only **0.763** (CV) / **0.714** (locked test) — a ~15–20 point generalization gap. The headline AUROC is yeast-inflated; the model became a strong yeast-RNA-LLPS predictor with moderate transfer to other organisms.

Figure 19 — v5 expansion: stability + specificity up, but the gain is yeast-concentrated

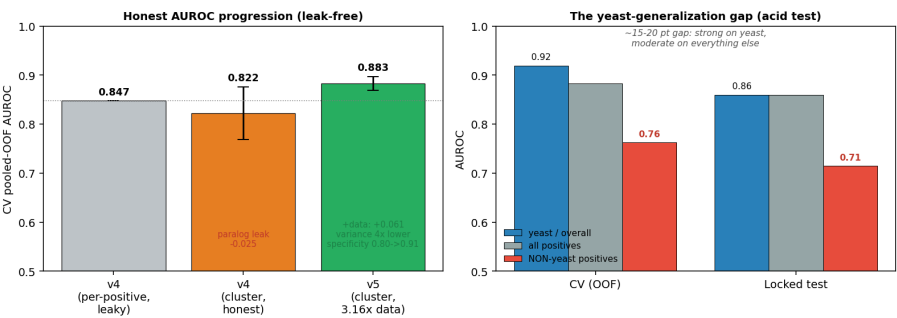


Figure 19. Left: the honest AUROC progression — correcting the v4 paralog leak (0.847→0.822) then the v5 expansion (→0.883, with 4× lower variance). Right: the yeast-generalization acid test — the model is strong on yeast (~0.92) but only moderate on non-yeast positives (~0.71–0.76), in both CV and the held-out locked test.

v5 verdict. 3.16× data bought **stability + specificity + yeast mastery**, but **not generalization** — that is gated by the field’s data monoculture and cannot be fixed with more (yeast) data. The remaining levers are **methodological**: organism-balanced training to stop the model over-fitting the yeast bloc, and tiling/multiple-instance learning to recover the 74%% of long-positive sequence currently truncated by RNA-FM’s 1,022-nt window. **v5 artifacts**: model/strict_eval_v5/final_model.pt, strict_pool_v5_positives.fasta (1,352).

9.4 Organism-balanced training (v6): closing part of the gap, for free

The v5 generalization gap (§9.3) is caused by the model seeing ~4× more yeast than non-yeast positives and optimizing for the majority. The direct methodological lever is to **rebalance the training sampler** so non-yeast positives are seen as often as yeast ones — per batch, **P(neg)=0.50, P(yeast-pos)=0.25, P(non-yeast-pos)=0.25** — with everything else (pool, cluster-grouped folds, seeds) held identical to v5.

Metric (5-fold cluster-grouped CV)	v5 (unbalanced)	v6 (org-balanced)
Overall OOF AUROC	0.883	0.884
Yeast-pos AUROC	0.919	0.909
NON-yeast-pos AUROC (n=268)	0.763	0.798
pos-vs-struct (specificity)	0.907	0.898

Organism-balancing is a **confirmed, near-free win**: non-yeast generalization rose **0.763 → 0.798** (+0.036 on n=268) while overall AUROC stayed flat (0.883 → 0.884) and yeast/specificity each slipped only ~0.01. The generalization gap narrowed ~30% (15.7 → 11.1 points). A first n=29 locked-test probe had misleadingly suggested a 0.033 overall cost; the n=268 pooled CV showed that cost was noise — the same ‘small held-out subsets lie, pooled CV tells the truth’ lesson as §6.5/§9.3. **Recommendation: adopt organism-balanced sampling in the production recipe.** It does not *close* the gap (11 points remain — the model still sees 4× more yeast), but it is a strict improvement at no cost.

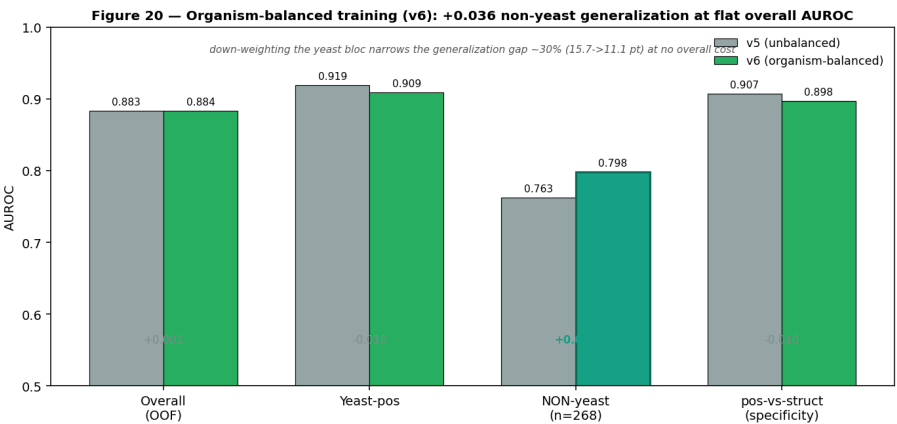


Figure 20. Organism-balanced training (v6) vs v5, both 5-fold cluster-grouped CV. The non-yeast generalization AUROC (highlighted) rises +0.036 at *flat* overall AUROC; yeast and specificity barely move. A near-free generalization gain from rebalancing alone.

v6 artifacts: model/strict_eval_v6_cv/.

9.5 Tiling / multiple-instance learning (v7): a different axis

998 of 1,352 v5 positives exceed RNA-FM’s 1,022-nt window and are truncated — ~74%% of positive nucleotides discarded. v7 recovers them with an **attention-MIL** model: each RNA is sliced into overlapping ≤1,022-nt windows (stride 512, ≤32), every window encoded by frozen RNA-FM + the adapter, then **attention-pooled over windows** (the model learns which windows carry LLPS signal), fused with the biophysics. Trained organism-balanced (stacking on v6) on the v5 pool.

MIL **works as a model** — it trained cleanly (inner-val 0.895, the highest of any variant) with comparable overall AUROC (locked test 0.852 vs v5 0.860). **But it does not close the generalization gap**: non-yeast AUROC came in at **0.685** (vs v5 0.714, v6 0.749) — not an improvement. This matches the mechanism: the long sequences MIL recovers are overwhelmingly yeast (median 2,530 nt), while the non-yeast positives are short

and already fully seen, so MIL enriches yeast/long-sequence representation, not cross-organism transfer. (All non-yeast locked-test numbers are $n=29$, so differences are within noise; the point is MIL clearly did not *help*.) **Conclusion: the two levers attack different axes** — organism-balancing is the generalization lever (§9.4); MIL is a viable architecture for proper long-RNA scoring (it doesn't discard 74% of the sequence) but not a generalization fix. v7 artifacts: model/strict_eval_v7_mil/.

10. Limitations & Next Steps

- **Pool size & the generalization ceiling (see §9).** The v5 expansion (3.16×) fixed the stability half of this limitation — fold variance dropped $\sim 4\times$ — but exposed the real ceiling: strict RNA-self-LLPS data barely exists outside one yeast screen, so the pool is $\sim 67\%$ yeast and the model generalizes to non-yeast RNA only moderately ($\sim 0.71\text{--}0.76$ vs 0.92 on yeast). More data cannot fix this. The open levers are methodological: **organism-balanced training** (down-weight the yeast bloc) and **tiling/multiple-instance learning** to recover the 74%% of long-positive sequence currently truncated by RNA-FM's 1,022-nt window; plus positive-unlabeled learning (the 'negatives' are really unlabeled) and multi-task auxiliary heads.
- **Composition ceiling.** The frozen-backbone model is composition-driven; the new absolute-repeat features + unfrozen last-2 layers (CV run) directly target the repeat-count and periodicity gaps the diagnostic exposed.
- **Species skew.** Human/mouse dominate; trace species (rice, *S. pombe*, xenopus, zebrafish) are too sparse for per-species evaluation and serve as out-of-distribution probes.
- **Context window.** RNA-FM caps at 1,022 nt; long scaffolds (e.g. NEAT1, ~ 23 kb) are handled by curating the mapped LLPS subregion or by sliding-window inference rather than switching backbone (which would break the $O(L^2)$ FEGS structure-injection).
- **CGG series caveat.** The verification gave the CGG repeat series internally inconsistent verdicts; flagged for human review before publication.
- **Kissing-loop OOD limit (v4).** The model cannot reject designed-nanostar scrambled controls because the kissing-loop condensation mechanism is absent from the natural-RNA training corpus (§8.3). Gaining this capability requires adding designed-nanostar RNAs (Fabrini/Takinoue-style KL positives + scrambled negatives) to training, together with a fresh independent external set.

11. Artifacts

- **Production model:** model/phase1/hybrid_best.pt (test AUROC 0.764).
- **Strict baseline:** model/strict/hybrid_strict_best.pt (AUROC 0.63).
- **5-fold CV:** model/strict_cv/ (running; per-fold checkpoints + cv_summary.json).
- **Datasets:** Data/raw/multispecies/strict_pool_{positives,negatives,hard_negatives}.fasta; unified_all_{positives,negatives}.fasta.
- **Audit trails:** strict_curation_audit.json (67 candidates), rebase_audit.json (21 leads) — every decision with PMID + reasoning.
- **Tools:** predict_hybrid.py (sliding-window inference), Functions/generator_hybrid.py (SeqProp de novo design).